AI Churn Prevention for a Telecom Operator

A data-driven retention proposal using real Client.csv and Record.csv outputs.

100,000	49.6%	0.689	78.3%
customers	churn rate	model AUC	top 10% risk churn

Final Assignment | Telecom Churn Prediction

Executive Summary

100,000	49.6%	78.3%
customers	churn rate	top 10% risk churn

- The dataset contains 100,000 customers and the observed churn rate is 49.6%.
- Gradient boosting performs best with validation AUC 0.689.
- The top 10% highest-risk customers churn at 78.3%, which is 1.58 times the average rate.
- The model should drive a tiered retention program: human contact for highest risk, low-cost outreach for medium risk.

Dataset Used

100,000	49.6%	0.689	78.3%
Rows	Churn	AUC	Top 10%

- Client.csv contains account profile, household, handset, credit and demographic variables.
- Record.csv contains usage, revenue, call quality, customer care and the churn target.
- The two tables are merged one-to-one by Customer_ID.
- Target variable: churn, where 1 means the customer left.

Current Churn Problem

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Rows	Churn	AUC	Top 10%

- Churn is close to one customer in two, so retention is a material revenue issue.
- Average monthly revenue is 58.7.
- Customers who churn show a larger decline in minutes of use: –22.8 versus –5.3 for customers who stay.
- Customers who churn have older equipment on average: 421 days versus 363 days.

Important Signals

- The strongest model signals are: months, eqpdays, change_mou, mou_Mean, avgqty, hnd_price.
- Usage decline is a useful warning sign before churn.
- Equipment age matters, suggesting handset upgrade offers may reduce churn.
- Some customer groups and regions have meaningfully higher churn rates.

Top Regions by Churn

Northwest/Rocky Mountain Area

56.9%

South Florida Area

53.4%

California North Area

52.1%

Modeling Approach

- Baseline model: logistic regression with imputation, scaling and categorical encoding.
- Main model: histogram gradient boosting to capture non-linear churn patterns.
- Validation method: stratified 80/20 train-validation split.
- Metric: ROC-AUC, which measures how well the model ranks churn risk.

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Model Output

- Logistic regression AUC: 0.631.
- Gradient boosting AUC: 0.689.
- Selected model: gradient boosting.
- The output is a churn risk score for each customer, not an automatic decision.

Selected Model

Gradient Boosting

AUC

0.689

Risk Tier Results

- Top 10% risk tier: 78.3% churn rate, 1,566 churn customers in validation.
- Top 20% risk tier: 72.9% churn rate, 2,914 churn customers in validation.
- This lift makes retention targeting more efficient than contacting customers randomly.
- Use top-tier scores for costly offers; use lower-cost messaging for wider groups.

Overall churn

49.6%

Top 10% risk

78.3%

Top 20% risk

72.9%

Lift in top 10%

1.58×

Threshold Simulation

- A broad threshold of 0.17 captures 99.8% of churn cases but targets almost all customers.
- Precision at that threshold is 50.0%, close to the base churn rate.
- Business interpretation: the threshold is useful as an early-warning flag, not as a universal discount rule.
- The practical policy should be tiered by score and customer value.

Confusion Matrix

TP 9,892 | FP 9,889

FN 20 | TN 199

Interpretation

Use tiers instead of one broad discount rule.

Retention Policy

- High risk + high revenue: retention specialist call and personalized offer.
- High risk + old handset: upgrade or device financing offer.
- High risk + usage decline: proactive service check and plan review.
- Medium risk: low-cost SMS/app campaign; low risk: monitor only.

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Financial Impact

- The campaign simulation estimates net value of 659,709 on the validation sample.
- Assumptions: 25% successful retention among true churn cases, six months of retained revenue and offer cost of 10 per contacted customer.
- The final business case should update these assumptions with actual campaign cost and customer lifetime value.
- The model gives a measurable way to compare retained revenue against offer spend.

Estimated Net Value

659,709

Update with real campaign cost before final pricing.

Operational Workflow

- 1. Score customers weekly or monthly using latest usage and service data.
- 2. Segment customers into high, medium and low churn risk.
- 3. Match action type to risk, revenue and churn reason.
- 4. Track acceptance, retained revenue and churn outcome for model refresh.

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Risks and Controls

- Do not treat the score as a final customer decision.
- Monitor churn rate, campaign ROI, contact rate and customer complaints.
- Avoid training on future information after churn is already known.
- Retrain after new campaign outcomes are collected.

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Quotation

- Churn scoring system implementation: 60,000.
- Monthly monitoring, retraining and reporting: 4,000.
- Optional performance fee: tied to verified retained revenue.
- The pricing is justified by recurring revenue protected through targeted retention.

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Recommendation

- Adopt the gradient boosting churn score as a retention decision-support tool.
- Start with the top 10% to 20% risk tiers because they have the clearest lift.
- Use the first campaign to measure true offer cost, success rate and retained revenue.
- Then retrain the model and scale the program.

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